Information Security Notes
Cryptographic Protocols and Authentication

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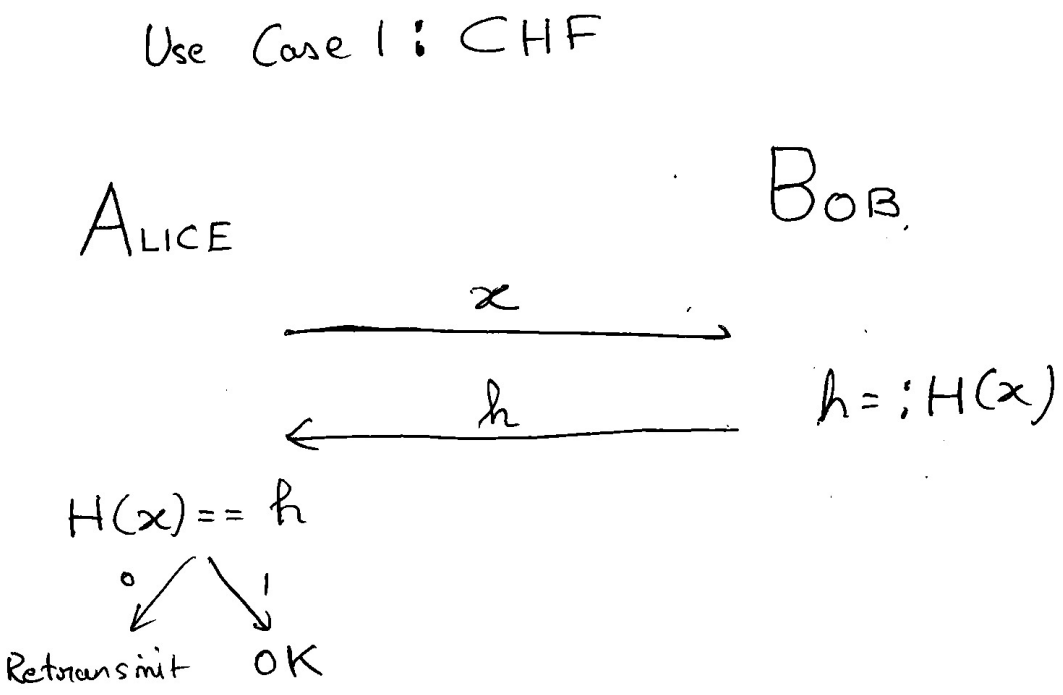
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1 Cryptographic Hash Functions (CHF)

1.1 Use Case:



Assumption:

1) There is No attacker in Between Alice and Bob

2) ~~Internet Routing protocols are very good.~~

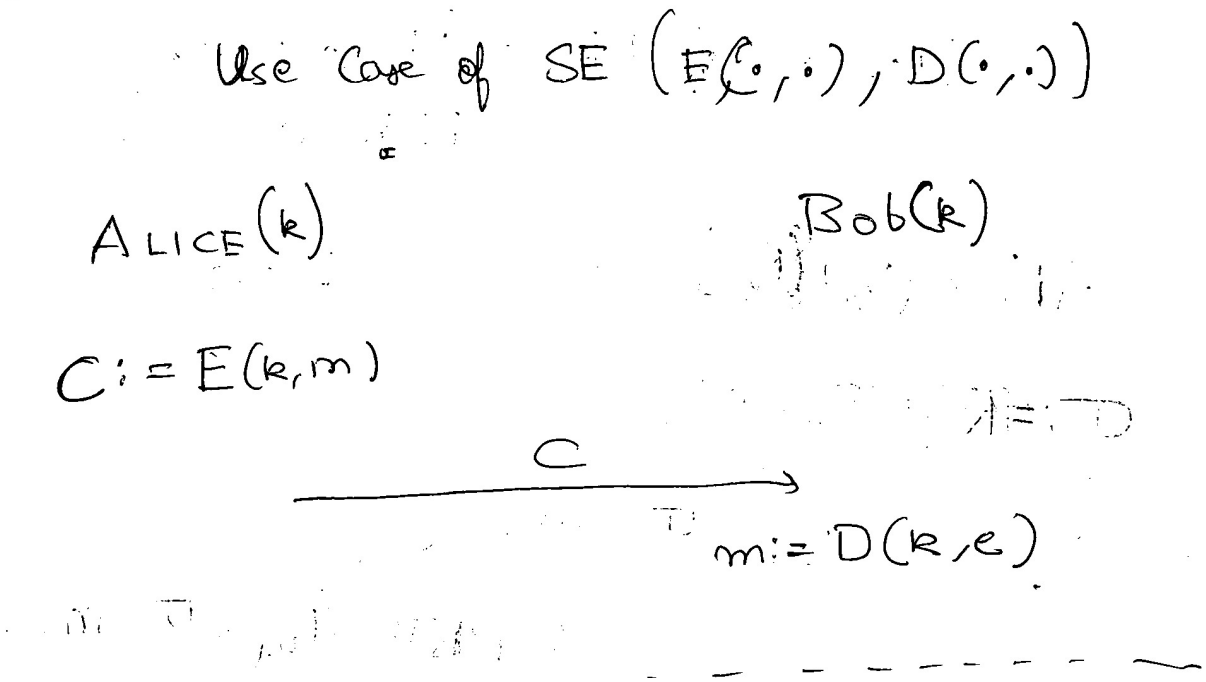
~~Only solves the problem of~~

2) Data can be flipped accidentally while in transit (This is the problem)

Hash function only solves this problem.

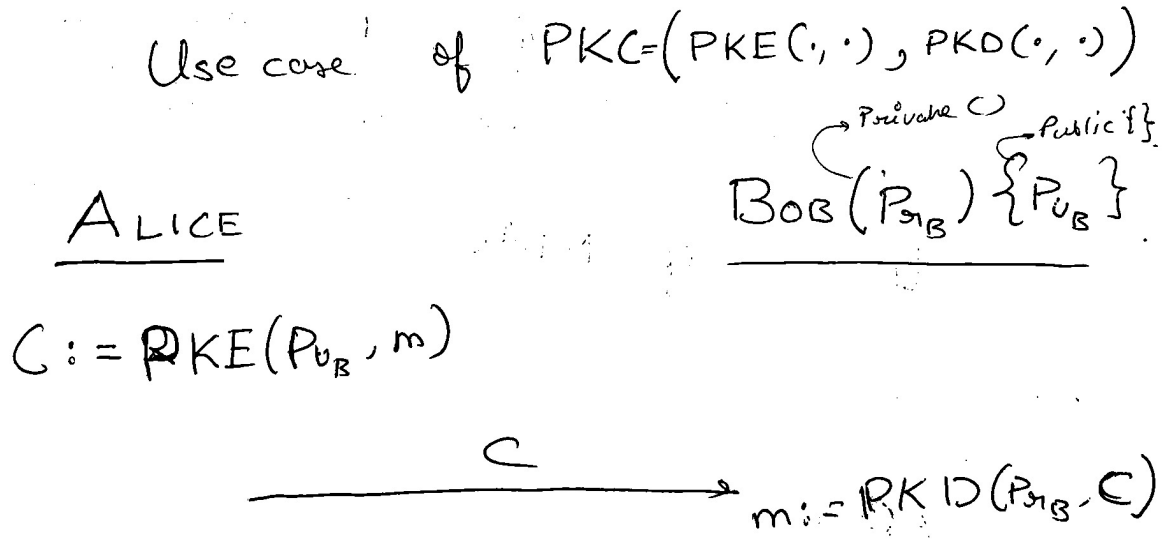
2 Symmetric Key Encryption (SE)

2.1 Use Case



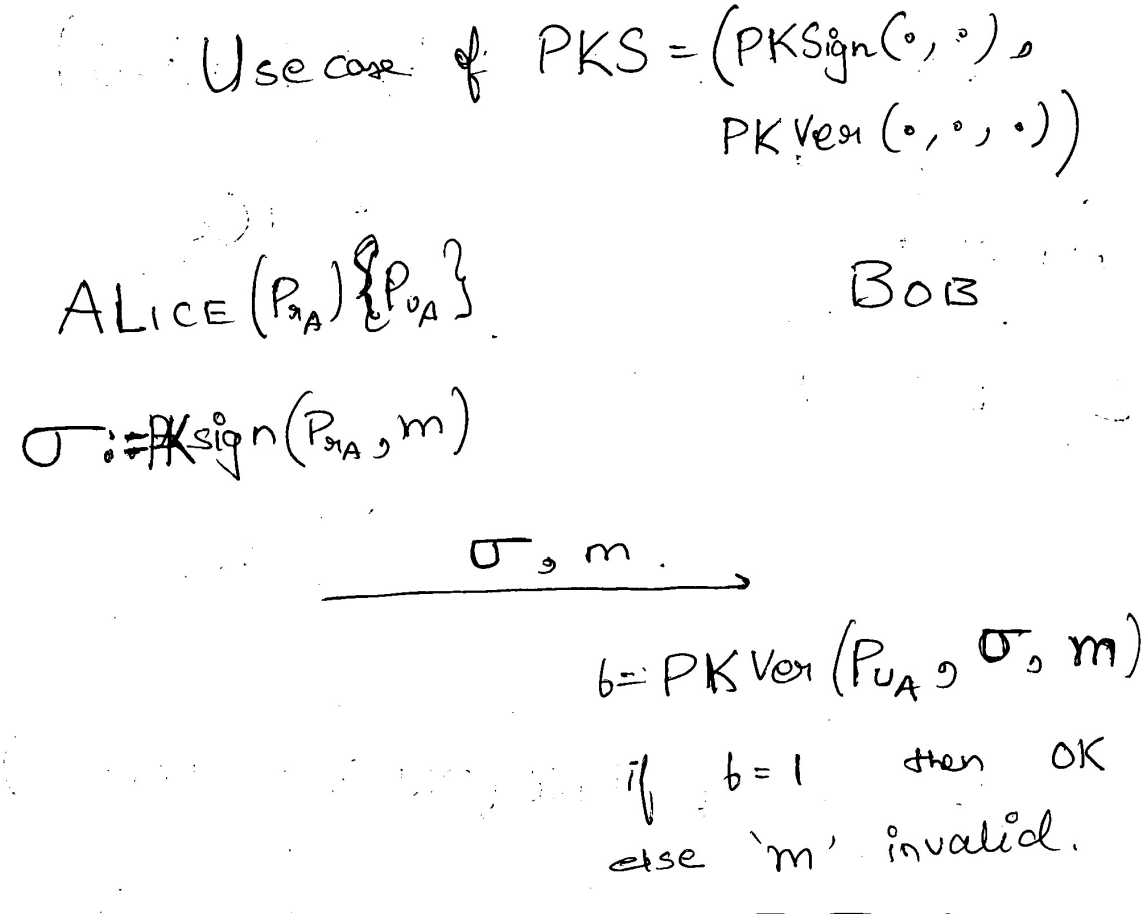
3 Public Key Cryptosystem (PKC)

3.1 Use Case



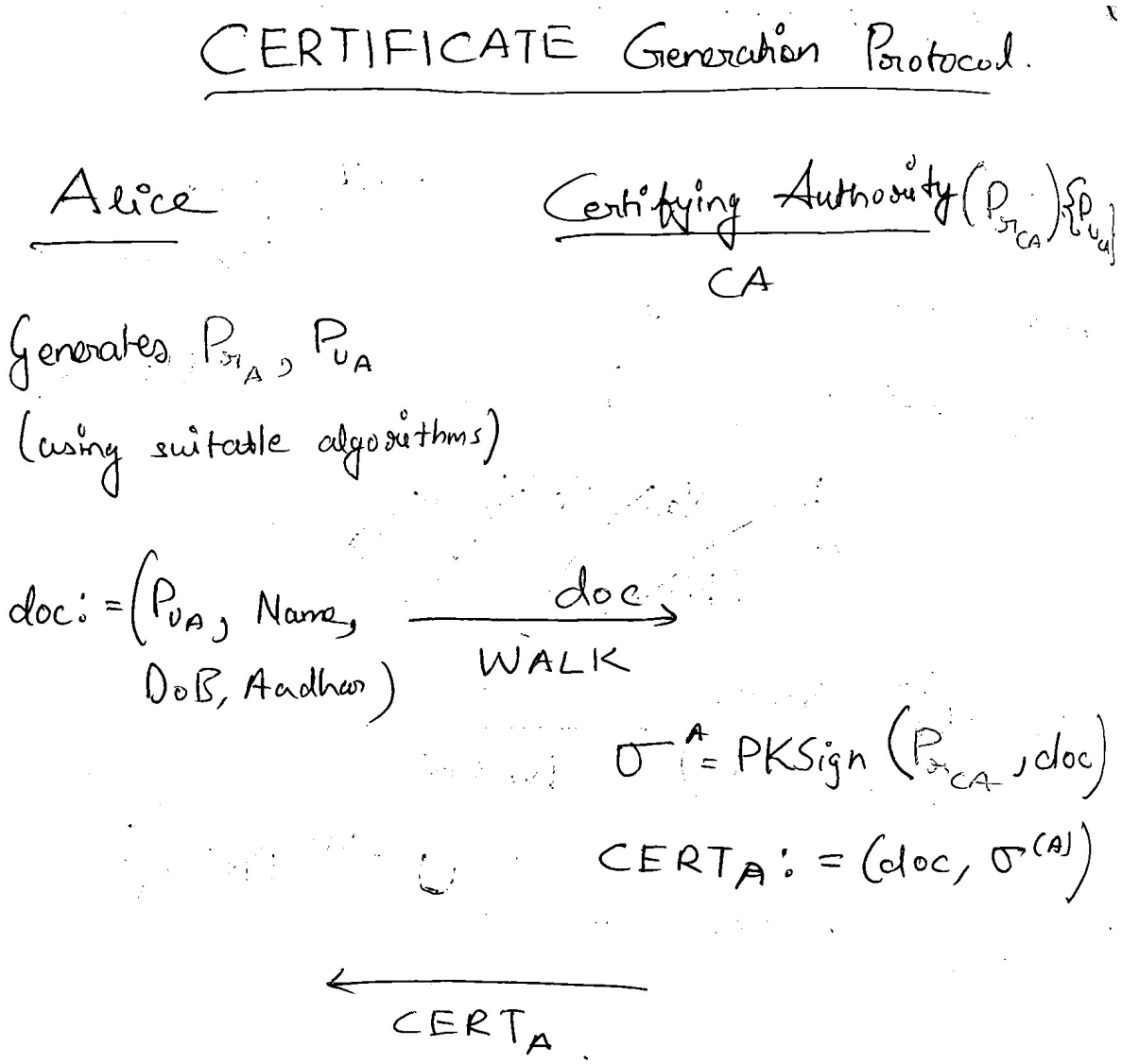
4 Public Key Signature Scheme (PKS)

4.1 Use Case



5 Certificates

5.1 Use Case



CERTIFICATE Verification Protocol

ANYBODY{$CERT_A$} BOB

$CERT_A$ →

- Extract $\sigma^{(A)}$ & doc from $CERT_A$
- Extract P_{U_A}, Name from doc.

$b := PKS\,Ver(P_{U_{CA}}, \sigma^{(A)}, doc)$

If $b == 1$
then P_{U_A} belongs to Name = "Alice"

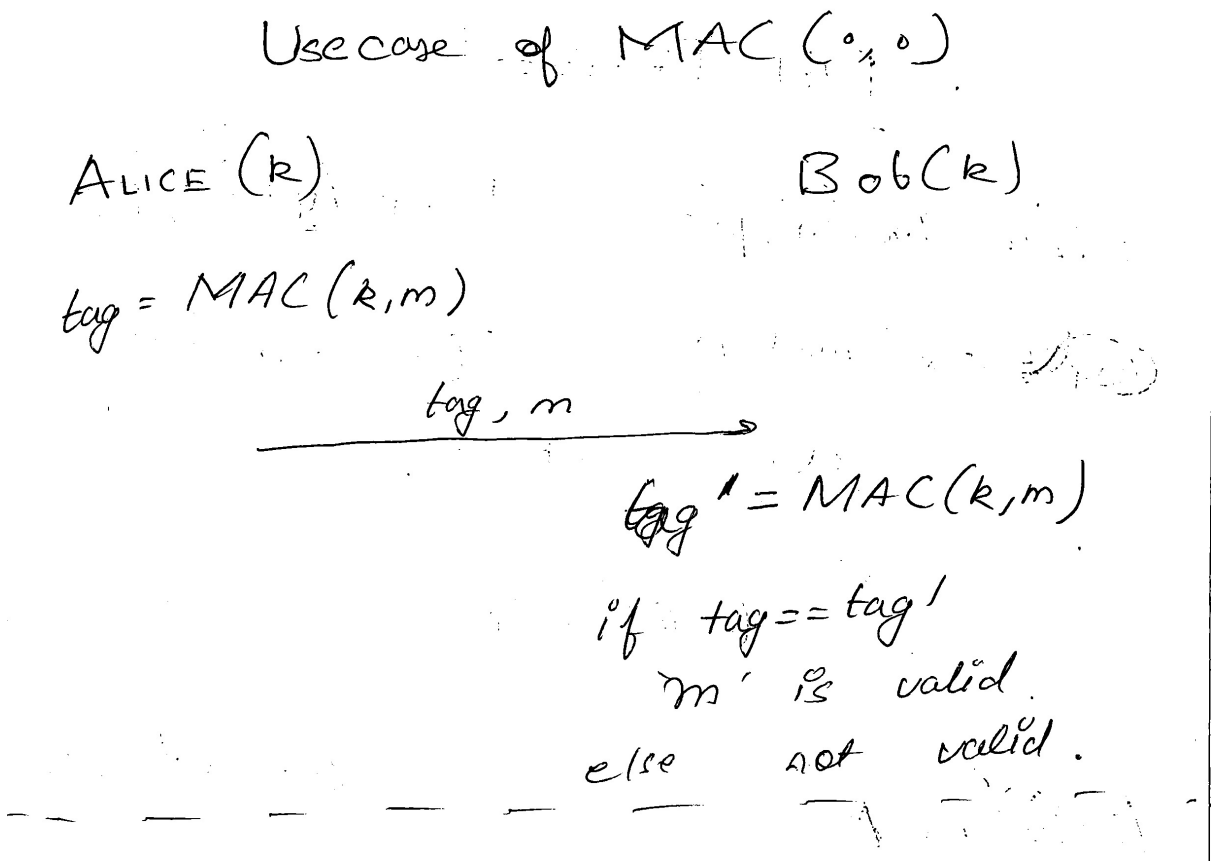
else
reject certificate.

Assumption:

1) Public Key of CA $P_{U_{CA}}$ → is known.

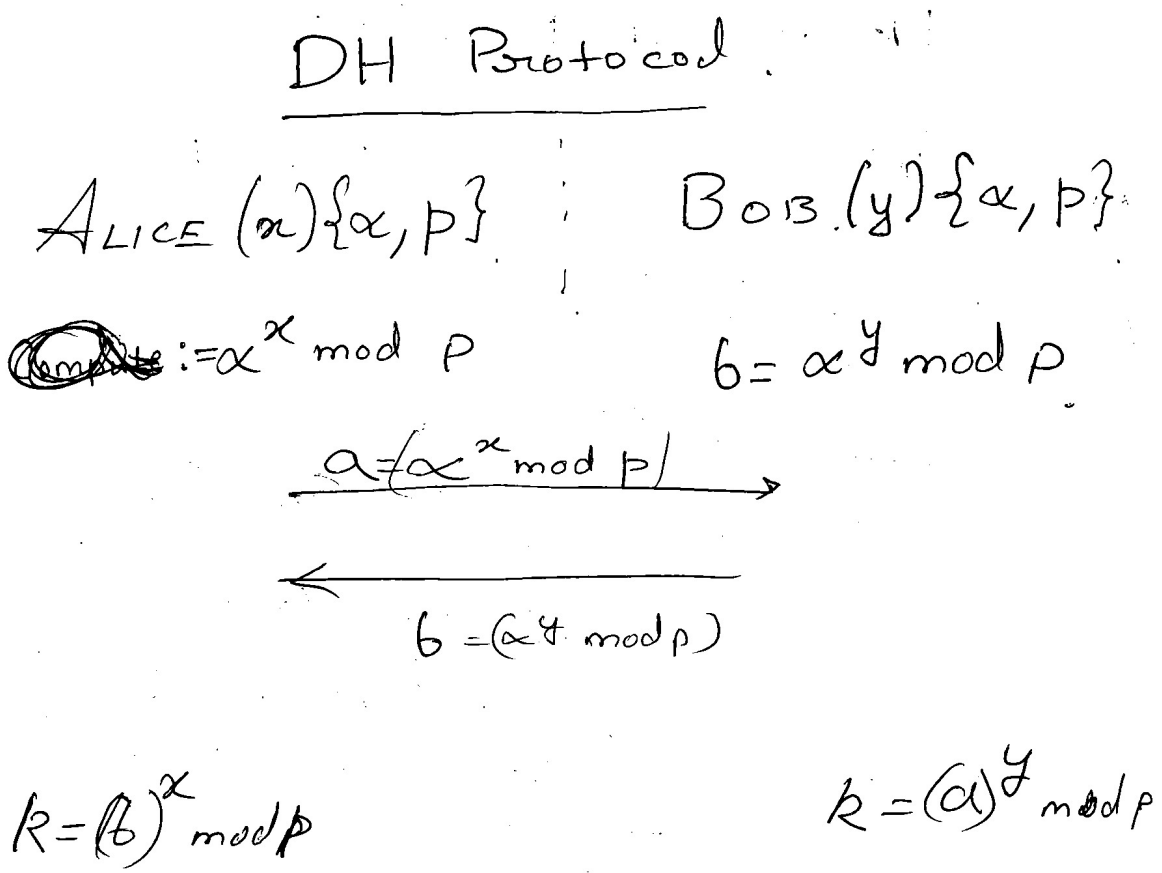
6 Message Authentication Code (MAC)

6.1 Use Case



7 Diffie-Hellman (DH) Key Exchange

7.1 Use Case



Assumptions:

1. Given α, α, p it is hard to compute x.
2. Given b, α, p it is hard to compute y.